TRINAY TELE-OPHTHALMOLOGY NETWORK

AI-ASSISTED RETINAL SCREENING CLINICAL DECISION-SUPPORT REPORT

CONFIDENTIAL MEDICAL DOCUMENT

DATE: 13 Oct 2026

DEPARTMENT OF RETINAL TELE-SCREENING & DIABETIC EYE CARE

1. SCREENING OUTCOME

Patient Name / ID:

Aviraj abasaheb Murkute (PX-001)

Age / Gender / Examined Eye:

20 Yrs / Male • Right Eye (OD)

Screening Operator & Date:

avi • 13 Oct 2026

DR Severity Prediction:

Moderate Non-Proliferative DR (Level 2)

ICDR Classification Staging

Model Score: 99.1%

Uncalibrated algorithm score; not a calibrated clinical disease probability.

Image Quality Assessment:

Good (Adequate)

Adequate capture clarity for automated triage; no repeat image capture indicated for quality reasons.

2. KEY AI FINDINGS

Supported Focal Lesion Findings & Landmark Localizations

PATHOLOGICAL FINDING	DETECTION STATUS	EXISTING MEASURED RESULT (RAW MODEL UNITS)
Retinal Microaneurysm (MA) (MA)	Detected	**345 px detected area (38 segmented regions)**
Retinal Hemorrhage (HE) (HE)	Detected	**2,075 px detected area (48 segmented regions)**
Hard Exudate (EX) (EX)	Detected	**4,611 px detected area (40 segmented regions)**
Soft Exudate / Cotton-Wool Spot (SE) (SE)	Detected	**269 px detected area (2 segmented regions)**

RETINAL ANATOMY LOCALIZATION:

Optic Disc: 4,528 px • Fovea: (245, 248) • Macula: Estimated ROI (r=56px) • Vessels: 12,901 px

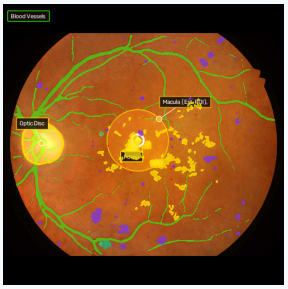
VISUAL EVIDENCE GALLERY

4 Canonical Model Inference Outputs (Proportionally Scaled Without Cropping)



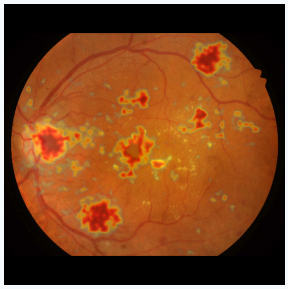
1. Original Fundus

True-colour reference



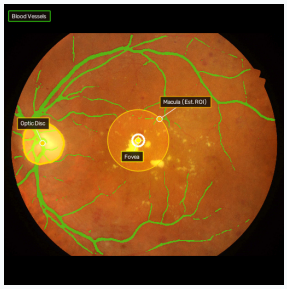
2. Combined AI Overlay

Lesions & landmarks



3. Grad-CAM++

Neural activation map



4. Anatomy Landmarks

Disc, fovea & vessels

3. CLINICAL INTERPRETATION

Status: Concordant (Moderate Non-Proliferative DR (Level 2))

Concordant Findings (Moderate Non-Proliferative DR (Level 2))

Whole-image neural classification (Moderate Non-Proliferative DR (Level 2)) is supported by focal lesion segmentation, which identified Microaneurysm, Retinal Hemorrhage, Hard Exudate, Soft Exudate (Cotton-Wool Spot) (total quantified footprint: 7,300 px detected area).

Model Analysis: Whole-image classification severity is concordant with localized lesion segmentation outputs.

All reported metrics reflect automated computer-vision model inferences and 2D segmentation masks. They do not constitute confirmed clinical diagnoses or replace in-person biomicroscopy.

4. RECOMMENDED ACTION

Prompt Specialist Review

Comprehensive Ophthalmology Referral (Prompt Evaluation)

Refer to a vitreo-retinal specialist for comprehensive dilated fundus examination, slit-lamp biomicroscopy, and macular optical coherence tomography (OCT).

Recommended Follow-Up Timing: Within 2 to 4 weeks (Prompt Specialist Review)

Guideline Basis: International Council of Ophthalmology (ICO) Guidelines for Severe / Proliferative DR

5. OPHTHALMOLOGIST REVIEW

Status: Pending Specialist Review

Reviewing Specialist: **Assigned Review Queue: Tele-Ophthalmology Network**

Review Date: **13 Oct 2026**

Clinical Assessment: **Pending specialist clinical counter-verification**

Assessment Notes: Queued for ophthalmologist counter-verification and review.

REVIEWING OPHTHALMOLOGIST:	MEDICAL REGISTRATION / LICENSE:	CLINICAL SIGNATURE & DATE:
Dr. _____	**MCI / Reg:** _____	**Date:** _____

CLINICAL DECISION-SUPPORT & REGULATORY DISCLAIMER

TRINAY is an artificial intelligence decision-support tool intended for tele-retinal screening and primary triage. It does not provide an autonomous medical diagnosis. All automated classifications, lesion segmentations, and landmark localizations must be clinically correlated and verified by a licensed ophthalmologist through direct clinical examination (slit-lamp biomicroscopy / OCT).